Templates in p_{IOI}
Then, [B] and [A] went to the [PLACE]. [B] gave a [OBJECT] to [A]
Then, [B] and [A] had a lot of fun at the [PLACE]. [B] gave a [OBJECT] to [A]
Then, [B] and [A] were working at the [PLACE]. [B] decided to give a [OBJECT] to [A]
Then, [B] and [A] were thinking about going to the [PLACE]. [B] wanted to give a [OBJECT] to [A]
Then, [B] and [A] had a long argument, and afterwards [B] said to [A]
After [B] and [A] went to the [PLACE], [B] gave a [OBJECT] to [A]
When [B] and [A] got a [OBJECT] at the [PLACE], [B] decided to give it to [A]
When [B] and [A] got a [OBJECT] at the [PLACE], [B] decided to give the [OBJECT] to [A]
While [B] and [A] were working at the [PLACE], [B] gave a [OBJECT] to [A]
While [B] and [A] were commuting to the [PLACE], [B] gave a [OBJECT] to [A]
After the lunch, [B] and [A] went to the [PLACE]. [B] gave a [OBJECT] to [A]
Afterwards, [B] and [A] went to the [PLACE]. [B] gave a [OBJECT] to [A]
Then, [B] and [A] had a long argument. Afterwards [B] said to [A]
The [PLACE] [B] and [A] went to had a [OBJECT]. [B] gave it to [A]
Friends [B] and [A] found a [OBJECT] at the [PLACE]. [B] gave it to [A]

Figure 14: Templates used in the IOI dataset. All templates in the table fit the 'BABA' pattern, but we use templates that fit the 'ABBA' pattern as well (i.e, by swapping the first instances of [B] and [A] in all of the above).

F Backup Name Mover Heads

Here we discuss in more detail the discovery of the Backup Name Mover Heads. As shown in Figure 15, knocking-out the three main Name Mover Heads surprisingly changes the behavior of the other heads that write in the IO-S direction (both positively and negatively). These heads compensate for the loss of function from the Name Mover Heads such that the logit difference is only 5% lower. We observe that the Negative Name Mover Heads have a less negative effect on logit difference, and 10.7 even has a positive effect on the logit difference after the knockout. The other heads that affected slightly positively the logit difference before the knock-out become the main contributors. Both the reason and the mechanism of this compensation effect are still unclear. We think that this could be an interesting phenomenon to investigate in future works.

Among the heads influencing positively the logit difference after knockout, we identified S-Inhibition Heads and a set of other heads that we called *Backup Name Mover Heads*. We arbitrarily chose to keep the eight heads that were not part of any other groups, and affected the logit difference with an effect size above the threshold of 2%.

In Figure 16 we analyze the behavior of those newly identified heads with similar techniques as Name Mover Heads. Those can be grouped in 4 categories.

- Four heads (9.0, 10.1, 10.10 and 10.6) that behave similarly to Name Mover Heads in their attention patterns, and the scatter plots of attention vs. dot product of their output with $W_U[IO] - W_U[S]$ (e.g 10.10 in Figure 16).
- Two heads (10.2, 11.9) that pay equal attention to S1 and IO and write both of them (e.g 10.2 in Figure 16).
- One head, 11.2, that pays more attention to S1 and writes preferentially in the direction of $W_U[S]$.
- One head, 9.7, pays attention to S2 and writes negatively.

We did not thoroughly investigate this diversity of behavior, more work can be done to precisely describe these heads. However, these heads are also the ones with the less individual importance for the task (as shown by their minimality score in Figure 7). The exact choice of Backup Name Mover Heads doesn't change significantly the behavior of the circuit.

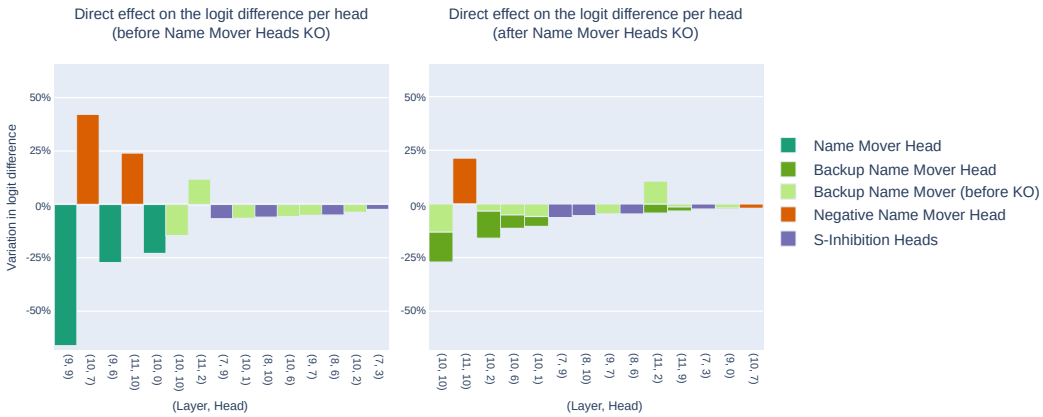

Figure 15: Discovery of the Backup Name Mover Heads. Left: results from the path patching experiment in Figure 3b. Right: the same path patching experiment results except computed after knocking out the Name Mover Heads. In both plots, the heads are ordered by decreasing order of the absolute value of their effect.

G GPT-2 SMALL FULL ARCHITECTURE

Here we define all components of the GPT-2 architecture, including those we don't use in the main text. GPT-2 small has the following parameters

- N: number of input tokens.
- V: length of vocabulary of tokens.
- d: residual stream dimension.
- L: number of layers.
- H: number of heads per layer.
- D: hidden dimension of MLPs.

It uses layer norm, the non-linear function

$$\text{LN}(x) \stackrel{\text{def}}{=} \frac{x - \bar{x}}{\sqrt{\sum_i (x_i - \bar{x}_i)^2}}, \tag{1}$$

where the mean and the difference from the mean sum are over all components of the dimension d vector in each sequence position. This is then followed by a learned linear transformation M (different for each layer norm).

In GPT-2 the MLPs all have one hidden layer of dimension D and use the GeLU non-linearity. Their input is the layer normed state of the residual stream.

We addressed the parametrisation of each attention head in the main text, and cover the technical details of the W_{QK} and W_{OV} matrix here: the attention pattern is $A_{i,j} = \text{softmax}(x^T W_{QK}^{i,j} x)$ where the softmax is taken for each token position, and is unidirectional. We then have $h_{i,j}(x) \stackrel{\text{def}}{=} M \circ \text{LN}((A_{i,j} \otimes W_{OV}^{i,j}).x)$.

Algorithm 2 describe how these elements are combined in the forward pass of GPT-2 small.

H VALIDATION OF THE INDUCTION MECHANISM ON SEQUENCES OF RANDOM TOKENS

We run GPT-2 small on sequences of 100 tokens sampled uniformly at random from GPT-2's token vocabulary. Each sequence `A` was duplicated to form `AA`, a sequence twice as long where the first and second half are identical. On this dataset, we computed two scores from the attention patterns of the attention heads:

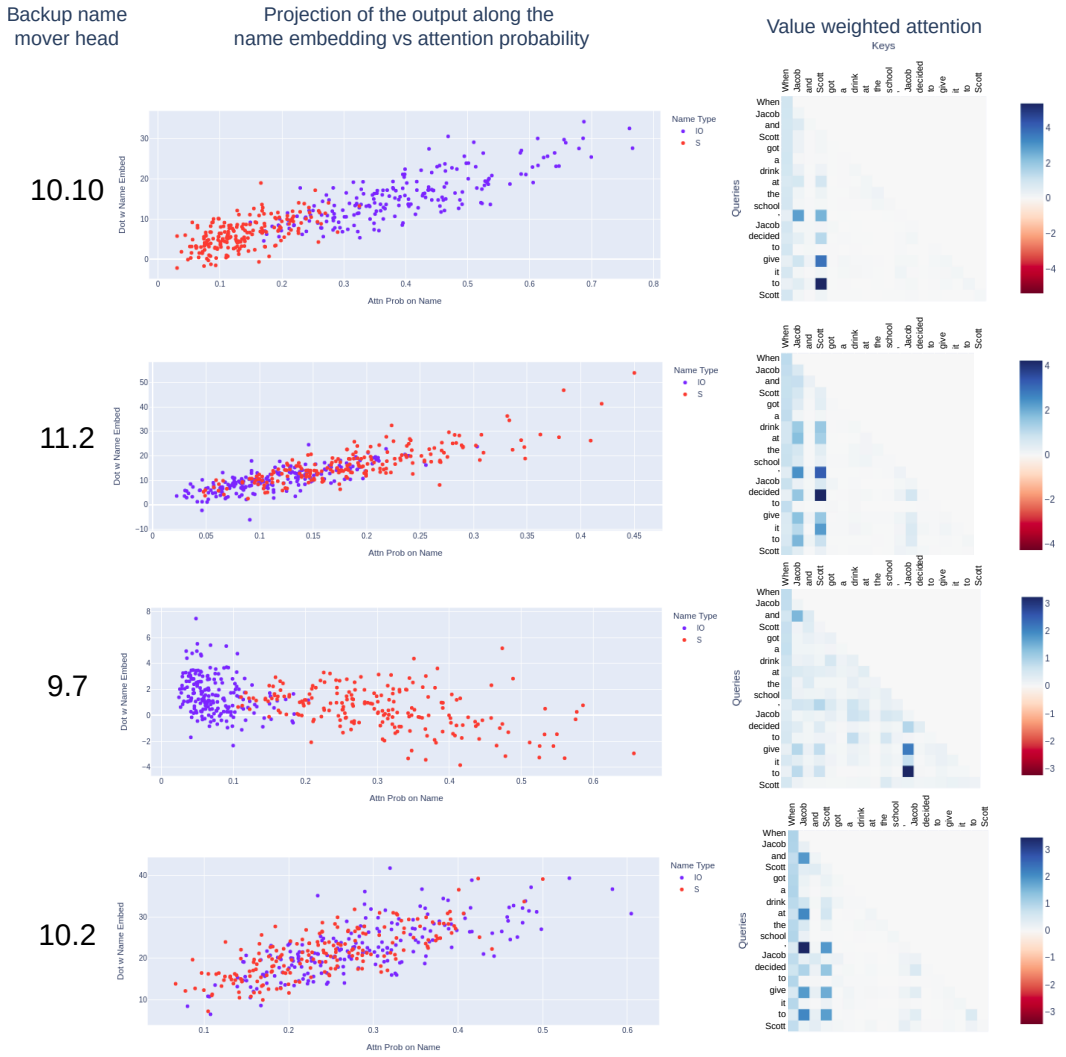

Figure 16: Four examples of Backup Name Mover Heads. Left: attention probability vs projection of the head output along $W_U[IO]$ or $W_U[S]$ respectively. Right: Attention pattern on a sample sequence.

Algorithm 2 GPT-2.

Require: Input tokens T; returns logits for next token.

$w \leftarrow$ One-hot embedding of T
$x_0 \leftarrow W_E w$ (sum of token and position embeddings)
for $i = 0$ to L **do**
  $y_i \leftarrow 0 \in \mathbb{R}^{N \times d}$
  for $j = 0$ to H **do**
    $y_i \leftarrow y_i + h_{i,j}(x_i)$, the contribution of attention head (i, j)
  end for
  $y'_i \leftarrow m_i(y_i)$, the contribution of MLP at layer i
  $x_{i+1} \leftarrow x_i + y_i + y'_i$ (update the residual stream)
end for
return $W_U \circ M \circ \text{LN} \circ x_L$